

Report about .Net version, Namespace, .Net core and Solution

.NET Versions Evolution:

- **.NET Framework (2002–2019):** Windows-only framework, tightly coupled with the OS (Versions 1.0 through 4.8.1).
- **.NET Core (2016–2019):** Complete redesign — cross-platform (Windows, Linux, macOS), open-source, lightweight, and modular (Versions 1.0 to 3.1).
- **Unified .NET (2020–Present):** Merged .NET Framework and .NET Core starting with .NET 5, moving forward to .NET 6, 7, 8, and beyond. Provides a single unified platform for Web, Mobile, Desktop, Cloud, and IoT.

Namespaces:

- Logical containers used to organize code elements (Classes, Interfaces, Structs) and prevent naming conflicts in large projects.
- *Example:* `System.IO` contains file manipulation classes, avoiding collision with custom classes named `File`.

Solution vs. Project:

- **Project (.csproj):** Contains the actual source code, assets, and dependencies required to build a single executable or library (.dll / .exe).
- **Solution (.sln):** A container/structure used to group and manage one or more related projects together.